

WASP-12b

R-1343

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_200326 · created 2026-07-31T21:50:48+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458934.68319 +/- 0.00072 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.152 +/- 0.044
Transit depth (Rp/Rs)^2	2.32 +/- 1.35 [%]
Orbital Inclination (inc)	86.73 +/- 2.87
Airmass coefficient 1 (a1)	0.45 +/- 0.31
Airmass coefficient 2 (a2)	0.34 +/- 0.23
Scatter in the residuals of the lightcurve fit is	66.47 %
Best Comparison Star	None
Optimal Aperture	1.3
Optimal Annulus	5.24
Transit Duration (day)	0.1001 +/- 0.0034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.152 ± 0.044 vs 0.1178 (deviation 0.67σ)
Duration fit vs published	0.1001 d vs 0.125 d (deviation 6.31σ)
Mid-time O–C	-0.2 min
Depth SNR	3.0
Residual scatter	66.47 %

Light curve

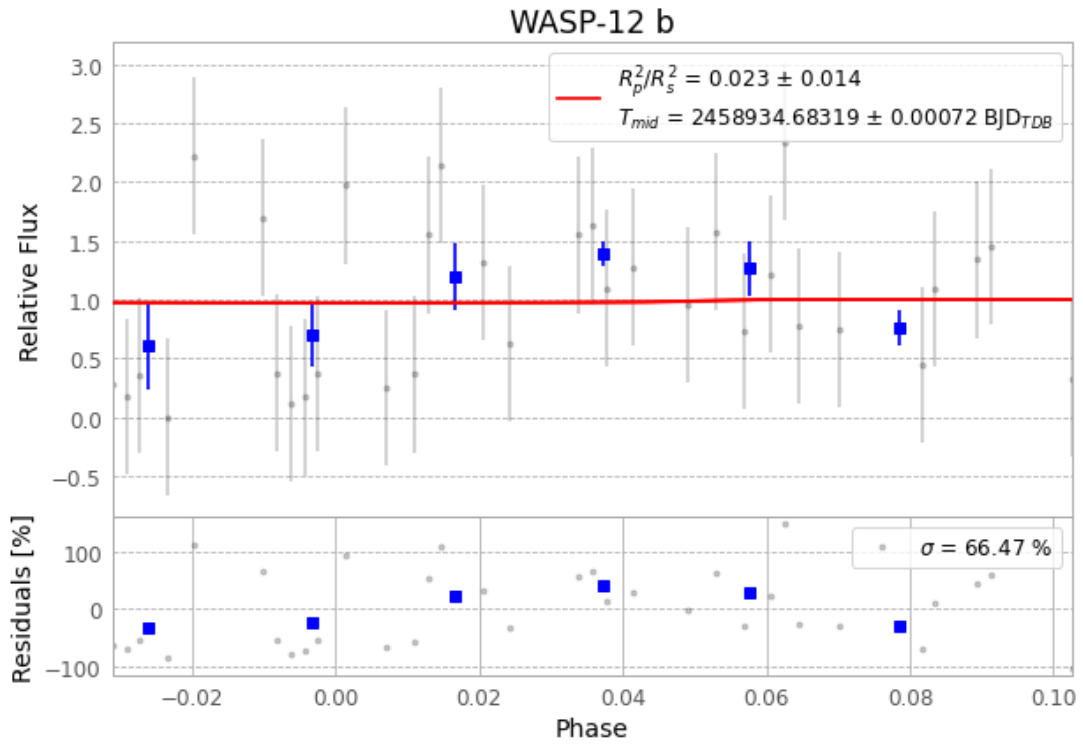


Figure 1 — FinalLightCurve_WASP-12 b_2020-03-25.png (SHA-256 924de597cfb51f24...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [261, 109], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ec03e5078168b2bb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

71 FITS frames (47 MB), WASP-12200326033312.FITS → WASP-12200326070313.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-200326.log (7325e68fcbda...), FinalLightCurve_WASP-12 b_2020-03-25.png (924de597cfb5...), FinalLightCurve_WASP-12 b_2020-03-25.csv (d886f89be908...)

Compute resources

Wall time 190.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 21:50Z.